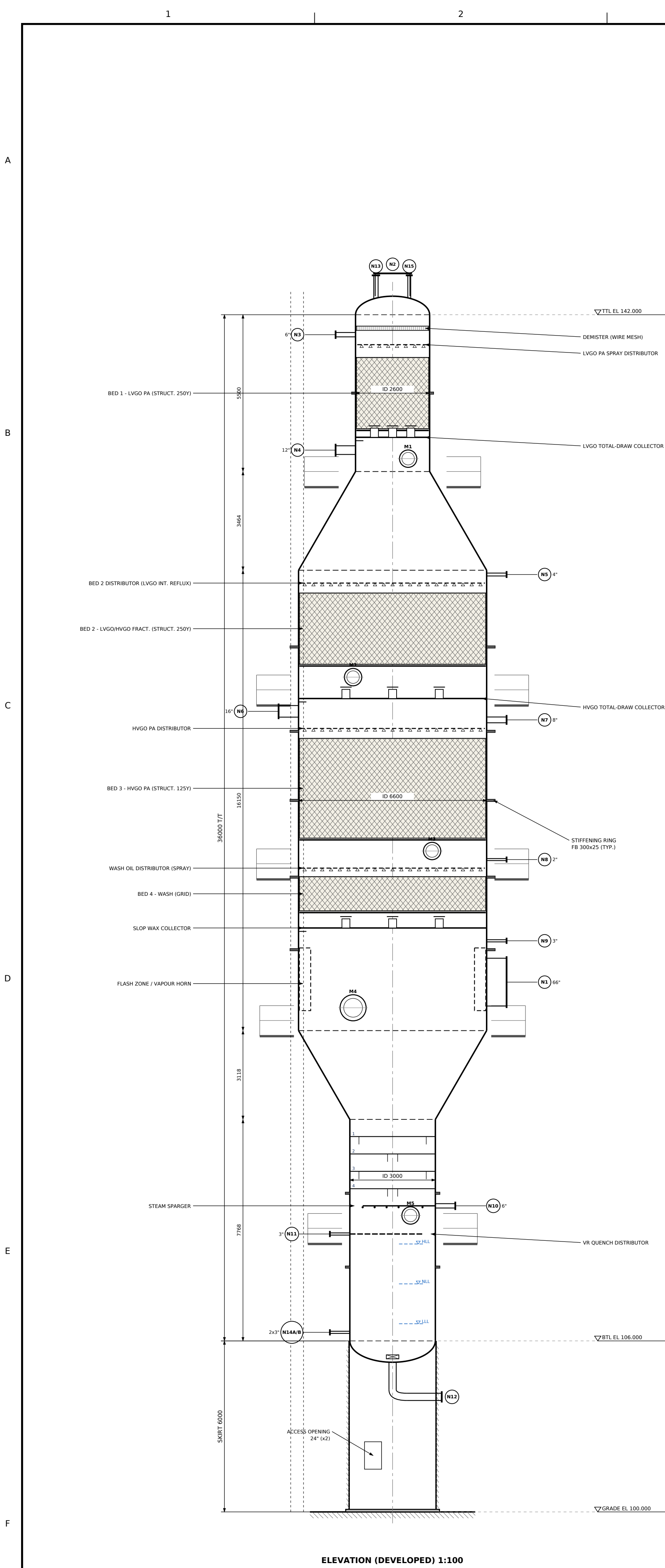




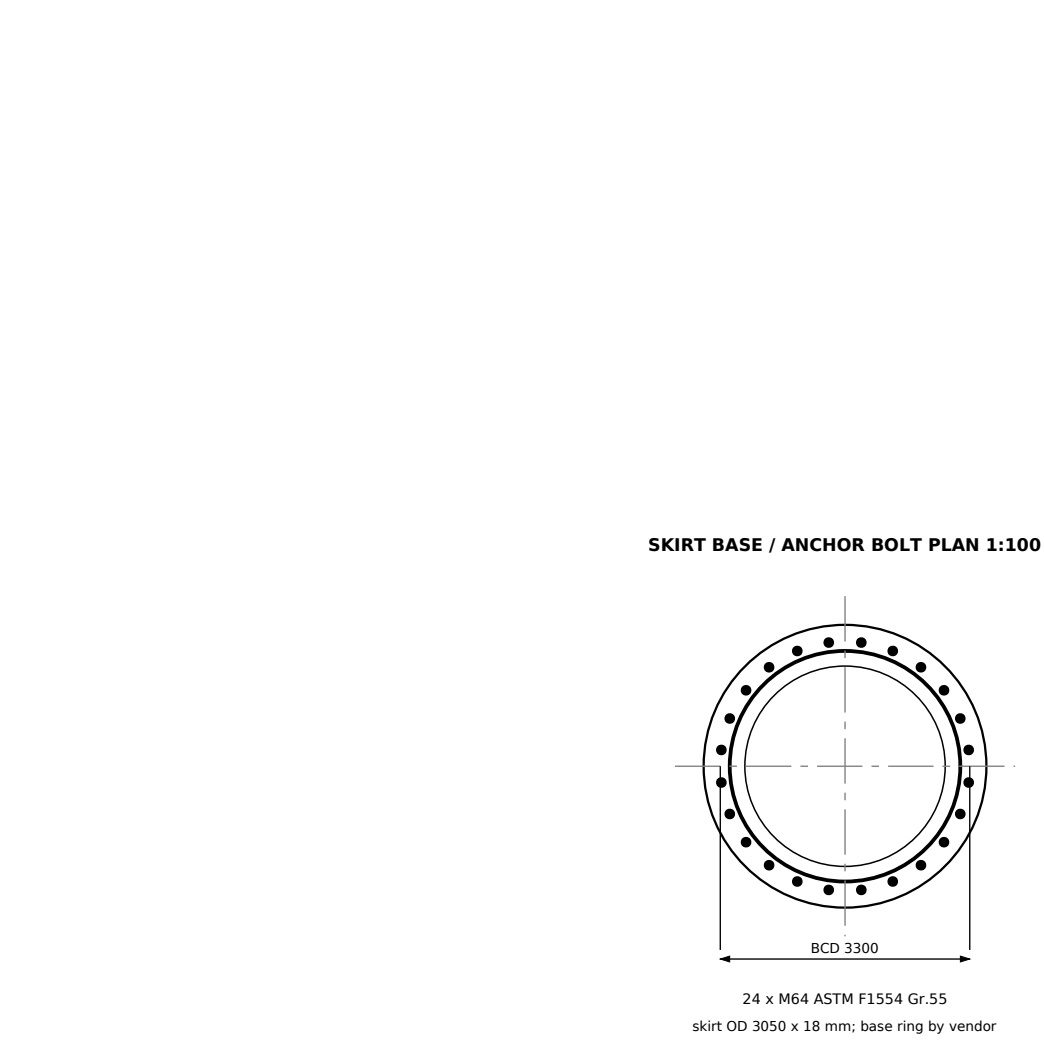
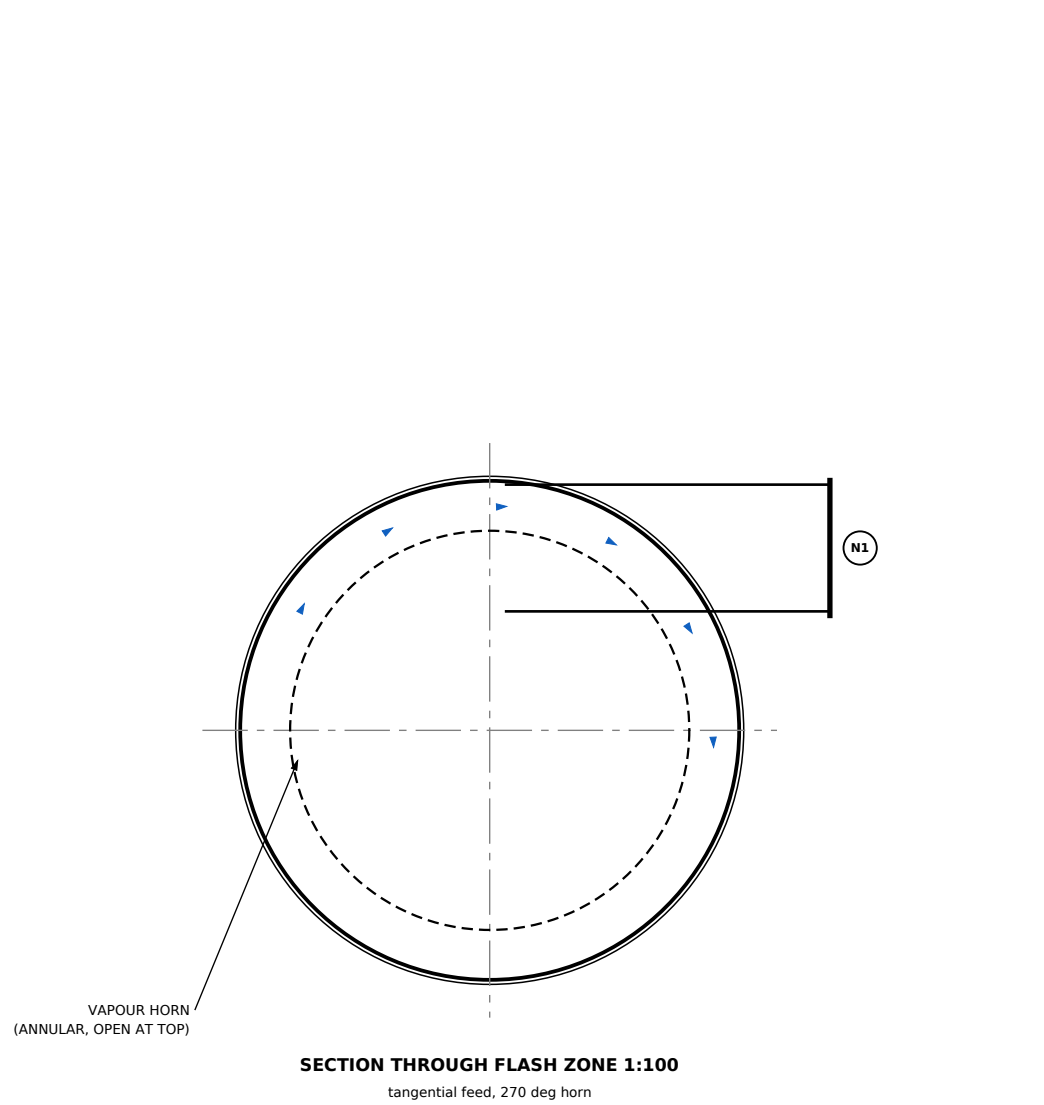
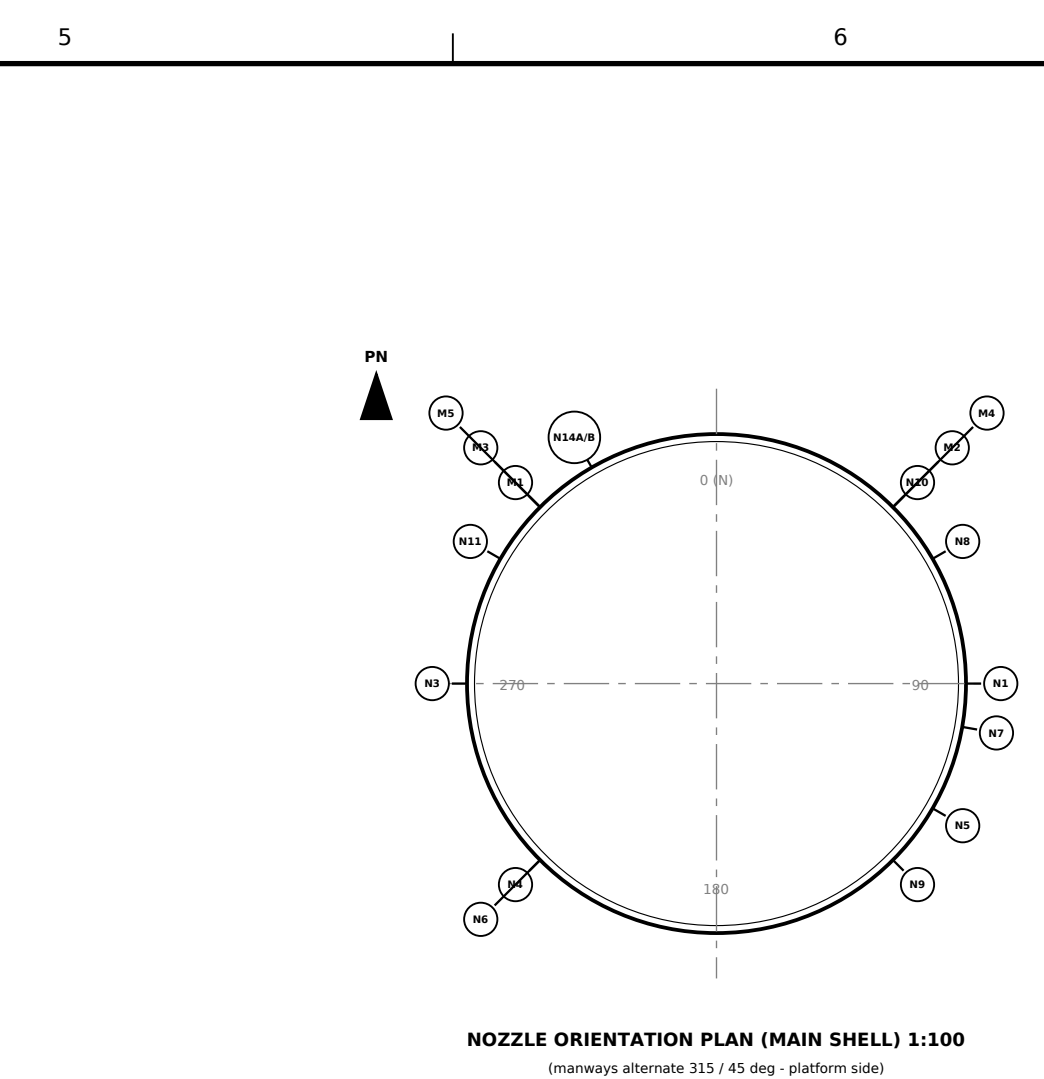
NOZZLE SCHEDULE						
MARK	QTY	SIZE	RATING	LOCATION	ORIENT	SERVICE
N1	1	6"	150# RF	EL 118.586	0	Feed - transfer line from H-201 (tangential to vapour horn)
N2	1	48"	150# RF	TOP HEAD	0	Overhead vapour to J201 E-202
N3	1	6"	150# RF	EL 141.300	90	LVGO PA return (to LVGO reflux)
N4	1	12"	150# RF	EL 137.250	225	LVGO total draw to P-201
N5	1	4"	150# RF	EL 132.886	120	LVGO internal reflux to bed 2
N6	1	6"	150# RF	EL 128.086	225	HVGO total draw to P-202
N7	1	8"	150# RF	EL 127.786	100	HVGO PA return
N8	1	2"	150# RF	EL 122.886	60	Wash oil (HVGO) to bed 4
N9	1	3"	150# RF	EL 120.036	135	Stop wax draw to P-203
N10	1	6"	150# RF	EL 110.738	45	Stripping steam
N11	1	3"	150# RF	EL 109.750	300	VR quench return (from E-201 outlet)
N12	1	10"	150# RF	8TH HEAD	0	Vacuum residue to P-204
N13	1	6"	150# RF	TOP HEAD	0	PSU-2001 relief (N)
N14#B	2	3"	150# RF	EL 106.300	330	LT / LG (slide boot)
N15	1	4"	150# RF	TOP HEAD	0	Vent / steam-out / N2 purge
M1	1	24"	150# RF	EL 136.950	0	Manway 24"
M2	1	24"	150# RF	EL 129.286	0	Manway 24"
M3	1	24"	150# RF	EL 123.186	0	Manway 24"
M4	1	36"	150# RF	EL 117.686	0	Manway 24" 36" in flash zone)
M5	1	24"	150# RF	EL 110.400	0	Manway 24"

HYDRAULIC SECTIONS (PROCESS SIZING)							
SECTION	V t/h	L t/h	rho V	u allow/f	D calc	D sel.	LOADING
Bed 1 - LVGO PA	10	83	0.037	16.26	2.49	2.60	92 % of Cs-limit
Bed 2 - LVGO/HVGO fract.	87	34	0.176	7.11	4.97	6.60	57 % of Cs-limit
Bed 3 - HVGO PA	178	161	0.283	5.39	6.41	6.60	94 % of Cs-limit
Bed 4 - Wash	152	13	0.307	5.77	5.50	6.60	70 % of Cs-limit
Flash zone	152	108	0.327	5.23	5.60	6.60	72 % of Cs-limit
Stripping	5	108	0.041	11.12	2.49	3.00	52 % flood

EXTERNAL PRESSURE UG-28/29 - FULL VACUUM (1.034 bar)							
COURSE	t nom	t ext	L / Le mm	Pa bar	RINGS	FB h x b	t no rings
Boot (stripping / quench)	25	18	3000	1.222	2	160x16	24
Lower cone 30 deg	25	25	2267	1.121	-	-	-
Upper shell (beds 2/4, flash zone)	25	24	3000	1.222	5	300x25	43
Upper cone 30 deg	25	25	2414	1.070	1	-	-
Top section (bed 1)	18	17	3000	1.209	-	140x16	20

SHELL COURSE / THICKNESS SCHEDULE						
COMPONENT	ID mm	FROM EL	TO EL	t mm	MATERIAL	REMARK
Skirt	3050 OD	100.000	106.000	18	SA-516-70	
Buttwn head 2:3	3000	106.000	106.000	18	SA-516-70	fireproofed
Booster (stripping / quencht)	3000	106.000	113.768	25	SA-516-70	
Lower cone 30 deg	6600 cone	113.768	116.886	25	SA-516-70	req 12.2 / ext 18
Main shell (beds 2-4, flss)	6600	116.886	133.036	25	SA-516-70	req 20.7 / ext 25
Upper cone 30 deg	6600 cone	133.036	136.500	25	SA-516-70	req 18.8 / ext 24
Top section (beds 1, 2)	2650	126.500	142.000	18	SA-516-70	req 20.7 / ext 25
Top head 2:1	2600	142.000	142.000	18	SA-516-70	req 11.3 / ext 17

PLATFORM SCHEDULE		
PLATF.	ELEVATION	SERVES MANWAY
P1	EL 109.400	M5
P2	EL 116.690	M4
P3	EL 122.190	M3
P4	EL 128.200	M2
P5	EL 135.950	M1
P6	EL 142.850	top head / davit



DESIGN DATA	
Service	Vacuum column (wet, packed)
Design pressure	3.5 barg / FULL VACUUM
Design temperature	415 °C
Operating P / T	20 / 60 mbara(e) / 70 / 386 °C
Shell material	SA-516-70
Cladding / lining	CS + 4105 clad (317L clad flash/wash zone)
Corrosion allowance	6.0 mm
Joint efficiency	1.0 (full RT)
Shell thickness	25, 25, 25, 25, 18 mm (by course, bottom up)
Heads (2:1 SG) top / btm	18 / 18 mm
Skirt OD x t x height	3050 x 18 x 6000 mm
Anchor bolts	24 x M64 on 3300 BCD
Hydrotest (top / btm)	6.91 / 10.56 barg
Insulation	125 mm mineral wool
Weight empty / oper.	294 / 337 t
Weight hydrotest	1044 t
Wind base V / M	649 km / 15641 kNm
Code	ASME VIII Div.1, U-stamp; ASCE 7-16

LEGEND	
	Valve tray with downcomer
	Liquid distributor
	Liquid level (LLL/NLL/RLI)
	Nozzle - mark and size
	Draw sump
	Packed bed (structured / grid)
	Collector / chimney tray
	Manway (shown on face)
	Platform with handrail
	External stiffening ring

WEIGHTS & LOADS	
Shell + heads + nozzles + clad	135.4 t
Skirt + base ring	10.8 t
Internals	97.9 t
Platforms / ladders	24.4 t
Insulation + fireproofing	25.9 t
EMPTY (installed)	284.3 t
OPERATING	337.5 t
HYDROTEST (full of water)	1043.8 t
Wind base shear / moment (ASCE 7, strength)	649 kN / 15641 kNm.
Natural frequency / gust factor	0.62 Hz / 1.22

- ## NOTES
1. Dimensions in mm, elevations in m. Grade = EL 100.000.
 2. Thicknesses are FEED minimum nominal (CFU-000-ME-CAL-01); fabricator to verify.
 3. Nozzle orientations preliminary - to be confirmed by piping study.
 4. Platforms 1200 wide, 60" w/rap, caged ladders not shown in full.
 5. Datasheet CFU-000-ME-DS-001: weights in data/mech.json.
 6. Design full vacuum: external stiffening rings at max. 3.0 m spacing (see table).
 7. Boot: VR quench distributor above HLL limits boot residence/cooking.
 8. HOLD: top section ID per equipment.json - see CAL-001 S13 (bed 1 rating).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
COLUMN GENERAL ARRANGEMENT			
C-201 VACUUM COLUMN			
DISCIPLINE	STAGE	SCALE	SHEET
MECHANICAL	FEED	1:100	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-200-ME-GA-004			A